AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

# TM276
# Software Development Processes and Methodologies

## Midterm Examination (MTA) - Makeup
## Spring Semester 2024/2025

## Date: 24 /Apr./2025

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| **Number of Exam Pages:** 3 (including this cover sheet) | **Time Allowed:** 90 Minutes |

**Instructions:**

- Total Marks: **60**
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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## Part I – Multiple Choice Questions — 6 Marks

*Answer **all** the following multiple-choice questions. Each question is worth 2 Marks.*

1. Which software process activity is where the customers and engineers define the software that is to be produced?
   a. Specification  b. Development  c. Validation  d. Evolution

2. Which principle of agile methods suggest that developers should wherever possible, actively work to eliminate complexity from the system?
   a. Customer involvement  b. Incremental delivery  c. Embrace change  d. Maintain simplicity

3. Which non-functional requirement metric can be measured by calculating the "Availability" of a system"?
   a. Speed  b. Portability  c. Ease of use  d. Reliability

## Part II -- Short Answer Questions — 24 Marks

*Answer **all** the following short answer questions. Each question is worth 8 Marks.*

1. Discuss "Acceptability" as an essential attribute of good software. Also discuss how essential acceptability is with respect to "Embedded control systems" and "Entertainment systems".

2. Briefly define what a prototype is.
   Give one usage for a prototype.
   List four benefits of prototyping.

3. a- Discuss what "ambiguous requirements" are and what problems might occur because of them.
   b- State why the following requirement is ambiguous and provide a possible correction by restating it properly without any ambiguity.

   "The system must limit the number of character that the user can use in a post"

## Part III – Problem Solving Questions — 30 Marks

*Answer **all** the following problem-solving questions. Each question is worth 15 Marks.*

1. Develop an activity diagram for the task "Returning an item" in a department store system assuming that the task can be completed by a salesman. Make sure to cover the following requirements:
   a. Use Swimlanes to represent 2 roles, the Salesman and the System.
   b. The Salesman will start the task and provide all the required information. The system will check the dates and will allow the return with a full refund if the elapsed time is less than 2 weeks, the system will allow the return with a store credit if the elapsed time is more than 2 weeks and less than a month, otherwise the return will be rejected.
   c. Make sure to use at least 8 activities.



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2. Consider an alarm system with the following description:
   a. The alarm has 3 states (Off, Ringing, Snoozed)
   b. The states will change based on 2 different timers
      i. The first timer will keep track of the On-Time and the Off-Time
      ii. The second timer will keep track of the number of seconds the alarm has been ringing.

   *Hint: when the alarm has been ringing for 30 second without any action, the state will automatically change to snoozed and the On-Timer will be incremented by 10 minutes.*

   Draw the state machine diagram for the alarm system representing the 3 states and at least 6 transitions and their respective events using Guards as you find necessary.

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**End of questions**

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